

Karan Kansal

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Software Engineer — Java, Spring Boot & Spring AI

EDUCATION

J.C. Bose University of Science and Technology, YMCA

Faridabad, India

Bachelor of Technology in Computer Engineering; CGPA: 8.38/10

2022 – 2026

Govt. Sr. Sec. School, Jhansa

Kurukshetra, India

Class XII; 94.6%

2021

EXPERIENCE

Yamaha Motor Solutions

Jan 2026 – Aug 2026

Graduate Engineer Trainee

Faridabad, India

- Designed and implemented **5+ backend modules** using **Spring Boot MVC**, enforcing separation of concerns and improving maintainability and testability of business logic.
- Built and integrated **15+ RESTful APIs** using Controller, Facade, Service, and Repository layers with **Spring Data JPA** for database persistence.
- Designed **10+ JPA entity models** mapped to **PostgreSQL** tables and implemented DTOs, Bean Validation, and centralized exception handling for consistent API responses.
- Developed responsive frontend features using **Angular and Vue.js**, integrating UI components with **Spring Boot REST APIs** across multiple application workflows.
- Collaborated with **5+ developers** across frontend and backend teams using **Git**, contributing to scalable full-stack application architecture and feature delivery.
- Tech Stack:** Java, Spring Boot, Spring Data JPA, REST APIs, PostgreSQL, Angular, Vue.js, Git

PROJECTS

PaperMind | *Java, Spring Boot, Spring AI, PostgreSQL, pgvector, React, TypeScript, JWT, Docker*

[GitHub](#)

- Built a **multi-tenant RAG application** using **Spring Boot** and **Spring AI** for document-grounded question answering with citations across PDF, DOCX, TXT, Markdown, and CSV files.
- Developed **REST APIs** for authentication, document management, semantic search, chat, and conversations using **Spring Data JPA**, DTOs, validation, and layered Controller–Service–Repository architecture.
- Implemented the RAG pipeline with **OpenAI**, **pgvector** similarity search, token-based chunking, history-aware query rewriting, asynchronous document ingestion, and **SSE** streaming for real-time AI responses.
- Built a **React + TypeScript** frontend and secured the full-stack application with **JWT + Spring Security**, enforcing per-user document/vector isolation and persistent conversational memory in **PostgreSQL**.

Vibe Coding Agent | *Python, LangGraph, MongoDB, Docker, Prompt Engineering*

[GitHub](#)

- Engineered a terminal-based AI coding agent using **LangGraph** with ReAct-style reasoning and dedicated planning, tool-execution, and reflection nodes for multi-step coding tasks.
- Enabled real-time streaming responses and **MongoDB**-based checkpointing for persistent, resumable sessions.
- Implemented guarded tool-calling for sandboxed file-system and package-manager operations to keep agent actions safe within the execution environment.

TECHNICAL SKILLS

Languages: Java, SQL, JavaScript, Python, C++

Backend & Frameworks: Spring Boot, Spring Data JPA, Spring Security, Spring AI, REST APIs, FastAPI

Databases & Tools: PostgreSQL, MongoDB, pgvector, Docker, Linux, Git/GitHub, AWS, Postman

AI & Agentic Systems: RAG, LangChain, LangGraph, LLM Agents, MCP, Prompt Engineering

Frontend: Angular, Vue.js, React, TypeScript

CS Fundamentals: Data Structures & Algorithms, DBMS, OOP, Computer Networks, System Design

ACHIEVEMENTS & CERTIFICATIONS

- Oracle Certified Associate (OCA), Java SE 8 Programmer (1Z0-808) — [Link](#)
- Solved **400+ Data Structures & Algorithms problems** across LeetCode, GeeksforGeeks, and Coding Ninjas